
A Deep Learning Multimodal Approach to Survival Risk Prediction in Kidney Cancer Patients

Hieu Nguyen
University of Chicago
nguyenhieu@uchicago.edu

Abstract

Cancer prognosis relies on heterogeneous data sources. To leverage the Cancer Genome Atlas (TCGA) data, deep learning models extract complex features and predict clinical outcomes. This work proposes a multimodal fusion strategy integrating clinical, RNA-seq, and histopathology data for survival prediction. Efficient embeddings and feature extraction models are constructed for each modality, followed by concatenation to optimize the negative Cox partial log-likelihood. Validated on TCGA's clear cell renal cell carcinoma dataset, the model achieves a high concordance index, primarily driven by clinical and RNA-seq features. This study lays the groundwork for future foundation model experiments and interpretability methods, such as attention-based heatmaps and integrated gradients, to enhance clinical relevance.

Code and trained model are made available at: <https://github.com/quanghieu31/multimodal-ccRCC>.

1 Introduction

Cancer prognosis via survival prediction aids biomarker discovery, patient stratification, and treatment response [1]. Understanding tumor microenvironments has improved prognosis [2], and advances in medical technology have enabled machine learning to analyze complex cancer data [3, 4]. Since prognosis depends on multiple factors, early risk identification is key, as smaller tumors generally indicate better survival and treatment feasibility.

Oncologists rely on both histology and genomic data, but histology often lacks genomic integration [1]. Whole-slide images provide morphological details but vary by pathologist, while genomic data offer molecular insights but cannot isolate tumor-specific transcripts. Fusing these modalities enhances early survival prognosis and personalized treatment [1], with deep learning well-suited for this task.

This study explores a deep learning approach with structured embeddings and feature extraction for survival risk prediction (Figure 3) in clear cell renal cell carcinoma (ccRCC), the most common kidney cancer [5]. The model integrates clinical, RNA-seq, and histopathology data, optimizing Cox partial likelihood and evaluating performance via the concordance index (c-index) and Kaplan-Meier curves. The study is expected to make appropriate rationale for embeddings in each modality and implement clustering and attention-based feature extraction for histopathology slides, with the view to stratifying the patient groups in high and low risk correctly.

2 Related Work

Cancer prognosis via survival prediction has gained traction due to the availability of cancer data. RNA-seq expression data have been used in regression models to assess gene contribution to sur-

vival, while other modern transcriptomic sequences accelerate the understanding of mutations and abnormalities [6, 7]. Multiple Instance Learning networks are recent developments that improve the feature extraction for whole slide images with efficient computation [8]. Another study develops a deep learning system to predict disease-specific survival across ten cancer types using histopathology images, employing a weakly supervised approach without pixel-level annotations and testing three survival loss functions [9]. It significantly improves risk stratification over traditional clinical variables, achieving high concordance index and retaining predictive value in multivariable analyses. Another one classifies non-small cell lung cancer types and predicts mutations on whole-slide images [10]. The traditional method often uses clinical features like tumor grades, staging, gender, or age, and fits into a Cox proportional hazards model [11].

Multimodal fusion via deep learning introduces more methods to experiment. It aims to understand the heterogeneity across data modalities, make discoveries and interpretations, and enhance human-computer interaction, especially in medicine. There are many ways to combine the modalities in this field. Natural language processing and convolutional neural networks are used to extract clinical features across ICU sources to predict clinical intervention [12]. Most works have focused on establishing correspondences between histology tissue and genomics: Pathomic Fusion model integrating histopathology and genomic features for cancer prognosis in glioma and ccRCC, correlation between gene expressions and cellular features or joint analysis of breast cancer histologic images and genomic covariates [1, 13, 14]. *TransSurv* is one of many prominent models that combine multi-scale pathological and multi-omics representations in a transformer-based mechanism to generate comprehensive, interpretable survival predictions [15].

In the related work that was reviewed, only a few explicitly described how their embeddings are derived in detail. This study aims to discuss the embeddings carefully and leverage deep learning, particularly in the histopathology feature extractions, to understand how to implement the models correctly in survival risk predictions.

3 Dataset and Embeddings

Three data modalities come from the TCGA-KIRC, hosted on the Genomic Data Commons Data Portal, National Cancer Institute [3]. A few patients with missing RNA-seq gene expression data are removed. Among the cancer datasets, TCGA-KIRC was selected since there are substantial data (533 patient-level samples) with full clinical information, RNA-seq expressions, and histopathology slides. Events are balanced in which two-third of the samples are alive and the rest are dead.

Variable distributions are in Figure 1. Most categorical clinical variables are imbalanced, which may lead to biased predictions. Only gender, age, staging, and tumor grade are selected as the clinical features used in this study. The first three features align with the choices made in the literature [1] but I added tumor grade since tumor grade and staging are not highly correlated and could contribute to the survival probability.

The age variable is standardized because it is normally distributed and has relatively large values compared to the other variables. Gender is binary encoded. Tumor grade and staging are one-hot encoded into binary representation for each category, ensuring no ordinal misinterpretation. It is also convenient since linear models (Cox proportional hazards model) or neural networks (fully connected layers) will assess each category independently, and we only have a few categories per tumor grade or staging (low cardinality). After pre-processing, there are 13 clinical features.

RNA-seq gene expression counts are high dimensional (19962 genes per sample). Only protein-coding genes are retained before normalization to reduce noise from non-coding RNA, ensuring biological relevant expressions [16].

The data from TCGA-KIRC also have several types of normalizations: raw counts, FPKM, or TPM. Initially, I had some confusion about the within-sample and between-sample normalizations. FPKM is not suitable for between-sample comparisons because the total number of FPKM normalized counts for each sample will be different [17]. For example, sampleA has a higher proportion of counts for XCR1 (5.5/1,000,000) than SampleB (5.5/1,500,000), despite identical FPKM values. Thus, XCR1 counts (or any gene) cannot be directly compared between samples due to differing total normalized counts. TPM is counts per length of transcript per million reads mapped that normalizes sequencing

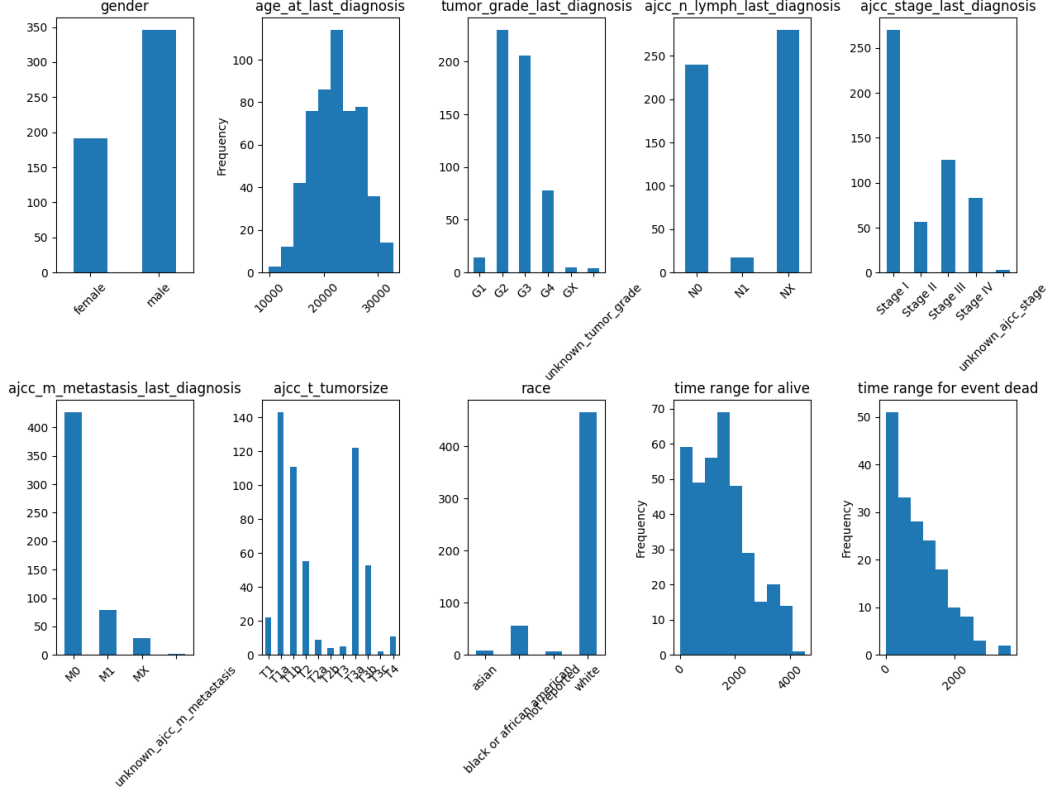


Figure 1: Clinical feature distributions

depth and gene length. TPM could be useful for between-sample comparisons. But since I want to start from a simpler normalization method first in this study, CPM (counts scaled by total number of reads) is used to apply to the protein-coding RNA genes to ensure that the gene matrices used in all analyses were the same [16, 17]. In particular, CPM of a gene count j for a patient i is,

$$CPM(i, j) = \frac{\text{rawcount}_{i,j}}{\text{totalreads}_i} \cdot 10^6$$

RNA-seq experiments generate different total read counts per sample, so dividing by total reads ensures gene expression values are comparable across samples. Then, often raw counts for each gene are small and so the ratio by total reads is also small, thus a multiplication of 10^6 avoids tiny decimal values. This scaling factor is also consistent with other popular normalization methods.

A log1P transformation is also applied on each gene to account large counts. Finally, since some patients have more one RNA samples, these will be average so that every patient has one single 19962-dimension vector of normalized gene counts.

As the final modality, each patient has a different number of the histopathology slides. The maximum number of slides being considered in this study is 5. Since the slides are large in size, the standard practice [10] is to get 20x magnification tiles (500x500) and 1000 random tiles are extracted from each slide. The choice for 1000 tiles is based on a few experiments to see which number gives not too dense and not too sparse tiles across slides on average. Also, the resolution level is 2 since my machine cannot read higher resolution (i.e. lower levels) due to memory issue. Then, a ResNet18 with ImageNet weights (the last classification layer is dropped) is used to convert these tiles into 512-dimensional numerical representations. ResNet18 is a common, state-of-the-art feature extraction model for image data, and its parameters are reasonable to run on my machine. ResNet18 helps detect the key features on each tile like the edges or color intensity.

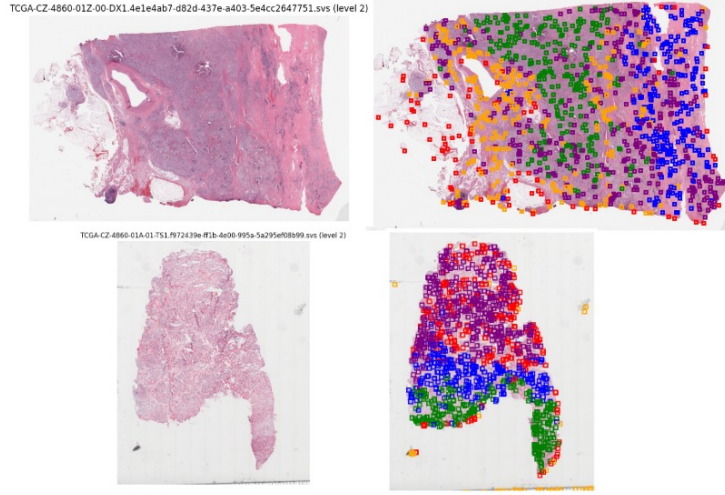


Figure 2: An example of clustering on 2 histopathology slides

Before inputting these features into the main model, a KMeans ($k = 5$) clustering method is used to group the tiles into 5 clusters. This approach was also done in [8]. The purpose is to localize the representations with an expectation that some clusters might be visually different than the others; that is, some clusters might observe tumor cells or complications. The choice for 5 clusters is based on intuition and would require more experiments to find the most optimal number.

Therefore, each patient’s histopathology slides can be represented as a list of 5 tensors, each tensor can have different shapes. For example, if a patient has 2 slides, which means 2000 tiles, then after clustering, they can have 5 tensors of shape (200, 512), (300, 512), (400, 512), (500, 512), and (600, 512) respectively. The clustering is applied across the total tiles of all the slides (Figure 2). That is, cluster 1 (blue) includes blue tiles from all the slides.

4 Methodology

4.1 Loss Function and Evaluation

The Cox proportional hazards model is a common semi-parametric approach for estimating the hazard function in survival analysis [11]. It assumes the proportional hazards which are satisfied in this dataset and that the hazard function can be parameterized as an exponential linear function $h(t|\vec{x}) = h_0(t)e^{\beta\vec{x}}$ at time t and with features $\vec{x}$ of a patient and learnable model parameters β , in which $h_0(t)$ is the baseline hazard that reflects how the risk of an event changes over time. It is hard to specify $h_0(t)$ for each patient, challenging the β estimation. Fortunately, the Cox partial log-likelihood (PL), without needing to specify $h_0(t)$, measures the relative probability of a patient’s risk against the other patients who are still censored [18]. Assuming these probabilities are independent between the patients, the PL of n patients in a batch with observed/uncensored events (dead, $D = 1$),

$$PL(\beta) = \prod_{i \in D=1}^n \frac{e^{\vec{x}_i^T \beta}}{\sum_{j \in R(t_i)} e^{\vec{x}_j^T \beta}}$$

in which $R(t) = \{j | t_j > t_i\}$ is the risk set (i.e. patients that are still censored at and after time t_i). Treating this as a loss function in a maximum likelihood estimation, it is more convenient to work with derivatives by turning it into a summation with logarithm and negation.

$$loss(\beta) = - \sum_{i \in D=1}^n \left(\vec{x}_i^T \beta - \ln \left(\sum_{j \in R(t_i)} e^{\vec{x}_j^T \beta} \right) \right)$$

Intuitively, the goal is to maximize the difference between the predicted risk score from the model, $\vec{x}_i^T \beta$, of an uncensored/dead patient and the total predicted exponential risk scores of the ones in

the risk set. Profiles of ones who pass away earlier should be considered at higher risk.

For model evaluation, c-index measures ranking accuracy - for any two patients, the one predicted to be at higher risk actually experiences the event before the other [19]. It is the percentage of all comparable pairs (at least one must be uncensored). Secondly, the Kaplan-Meier is used to plot the survival probability over time for two groups divided at the predicted median risk [20].

4.2 Model Architecture

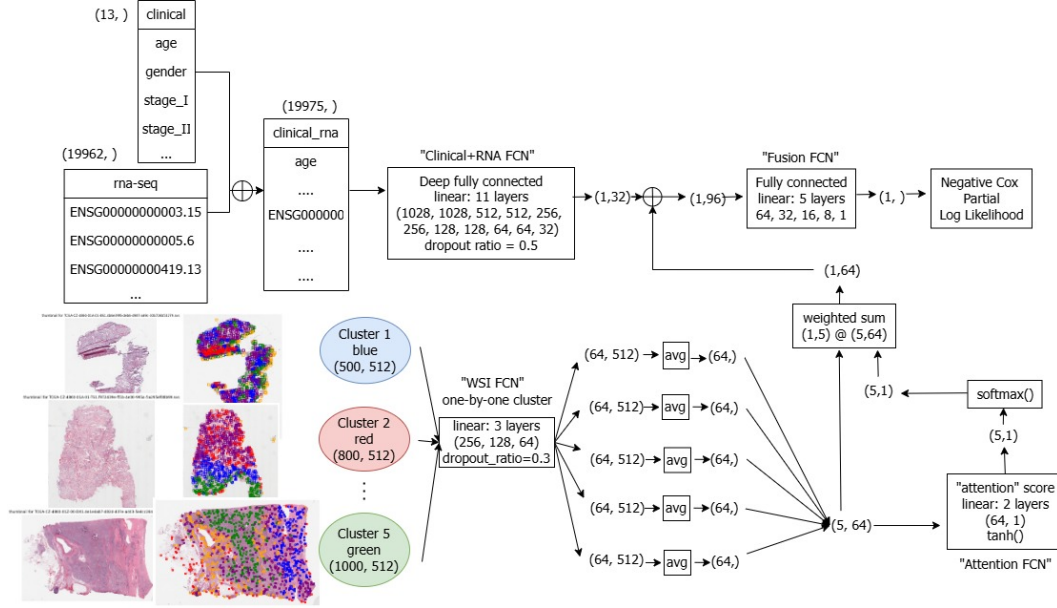


Figure 3: Model overview. "FCN" means fully connected network.

In Figure 3, the first part of the model is a simple concatenation of the processed clinical and RNA-seq expression features. The concatenated features are 19975 in dimension while there are 533 samples which is a relatively smaller number. Thus, overfitting is the primary issue to consider.

A deep fully connected neural network with heavy regularization (Clinical+RNA FCN) is a solution. Deep networks can learn hierarchical feature representations, while regularization techniques mitigate overfitting by enforcing sparsity and reducing reliance on specific features [21]. In this case, dropout ratio is useful because it prevents co-adaptation of neurons due to the deep fully connected layers [22]. By randomly dropping units, dropout reduces the risk of overfitting to the small training set for better generalization. Furthermore, gradually lowering the dimensions in each layer helps in progressive feature extraction, ensuring that the network captures essential information while filtering out noise. This structured reduction also aims to improve generalization by preventing the network from memorizing small training samples and learn lower-dimensional and more meaningful feature representations. For the clinical and RNA-seq data, the output dimension is 32.

In extracting the lower-dimensional features from the histopathology images, a notable thing is that the clustering is different between the samples. That is, cluster 1 of a sample and cluster 1 of another sample can refer to different characteristics of its corresponding set of tiles. The hypothesis is that the fully connected linear model for whole slide images (WSI FCN), through attention-based training, will finalize its weights to effectively reduce the 512 to 64 dimensions for each cluster while learning the differences between the clusters, especially, for each sample. In particular, WSI FCN takes one cluster at a time, condenses it to 64 dimensions and applies an average pooling, before stacking the 5 64-dimensional clusters into a small Attention FCN with Tanh activation and then softmax to calculate the weights for each cluster. Intuitively, the goal of these two models is learn the

difference between the clusters and how to place appropriate weights on each of them rather than deterministically indicating a cluster, say, is a tumor or not. Instead of rigidly categorizing clusters (i.e., tumor vs. non-tumor), the model learns to weigh clusters adaptively, preserving sample-specific feature representations. This hypothesis can also be better addressed by using an integrated foundation model, which is discussed in the Conclusion and Lessons Learned section.

To sum up, the Attention FCN model outputs 5 scores that are transformed with softmax to determine the probabilities being applied to each of the 5 clusters. The final output of the histopathology images is a tensor of shape 64.

The final concatenated features have the dimension of 96 before inputting to the Fusion FCN model. The model fuses the three modalities to estimate the loss function given the predicted risk score. The model is small because all of the heavy and complex computations are expected to happen in the other feature extraction models. The Fusion FCN aims to be simpler and captures the interactions of the modalities.

4.3 Experiment Set-up

The training started with some sets of conventional hyperparameters and regularization. This must be constrained by the number of samples and features in training. Training, validation, and test sets are of the 70/15/15 ratio. The experiments contained combinations of batch size of (32, 64), epochs of (5, 10, 15), learning rates of (0.001, 0.0001), weight decays of (0, 0.0001) and several attempts to change the dimensions of layers and number of layers in the Clinical+RNA FCN and WSI FCN models.

5 Results

The hyperparameters that give the highest c-index include batch size of 32, epochs of 5, weight decay of 0.001, and learning rate of 0.0001. The dropout ratios are 0.5 for the Clinical+RNA FCN and 0.3 for the WSI FCN model. In neural Clinical+RNA FCN, it seems that more hidden layers and dimensions per layer are correlated with high validation c-index (holding others constant).

The multimodality model is compared to 3 baseline unimodal models. The clinical baseline is based on the age, gender, tumor grade, and staging covariates, as discussed in Section 2. They are fitted into a CoxPH model with a $l1$ penalizer of 0.01 due to the sparsity and high multicollinearity. The RNA-seq gene expression baseline involves a Principal Component Analysis (PCA) with 16 principal components that are reduced from 19962 genes. They are fitted into a CoxPH model without any regularization. Then, a baseline model on the histopathology slides utilizes the same embeddings as in the multimodality model and the same negative Cox partial log-likelihood function. That is, this baseline fits ResNet18 outputs with KMeans ($k = 5$) clustering along with the WSI FCN and Attention FCN feature extraction models. It has 5 epochs, learning rate of 0.001, and batch size of 32. Another simpler model could be used to evaluate this baseline and will be done in the near future. Table 1 summarizes the models and the c-index results on test sets.

Model	c-index (std)
Baseline - Clinical	0.7631 (0.0514)
Baseline - RNA-seq	0.7326 (0.0410)
Baseline - Histopathology	0.6868 (0.0560)
Multimodality	0.7753 (0.0427)

Table 1: C-index results. Mean c-index is calculated on test sets with 300 bootstraps with replacement.

All models have good results. The baseline model for histopathology slides has the c-index that is relatively lower than the other three. Meanwhile, the multimodal model has the highest c-index. Clinical and RNA-seq features likely contribute most significantly to the multimodal model’s high c-index. It implies that the histopathology might play a small part in the multimodal effects.

Based on these c-index results only, the approach of combining clinical, genomic, and histopathology data outperforms unimodal models although the histopathology features alone do not add significant improvement over the clinical or RNA-seq gene expressions.

Nonetheless, the integration can distinguish survival Kaplan-Meier curves well (Figure 4). We observe that the multimodality version allows for more fine-grained stratification of survival probabilities between the low and high risk groups. This information can be used to understand risk cohorts and develop personalized treatments. Meanwhile, other than the RNA-seq baseline model, the other three

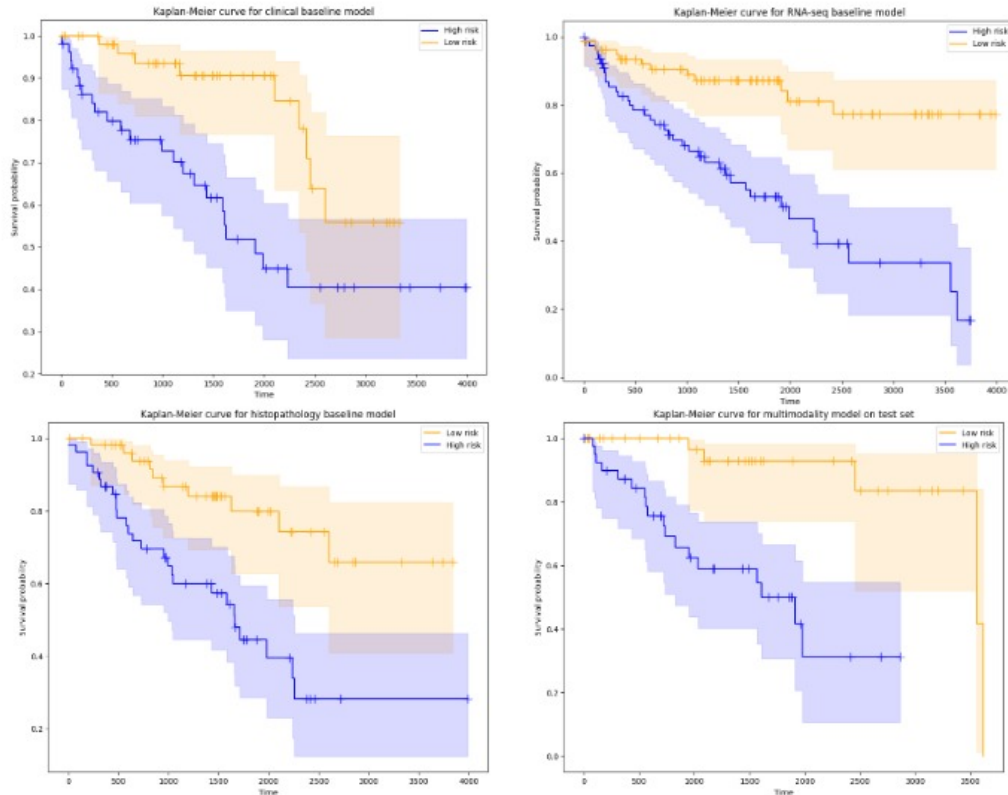


Figure 4: Kaplan-Meier curves

Kaplan-Meier curves have wide confidence intervals. This might indicate a high variance among the two risk groups that needs to be investigated further in the future.

6 Conclusion and Lessons Learned

This study demonstrates that integrating clinical data, RNA-seq gene expressions, and histopathology images improves survival risk prediction compared to unimodal approaches. The multimodal fusion model achieved the highest concordance index, with clinical and RNA-seq features contributing most significantly. While histopathology images added some predictive power, their effect was relatively minor, suggesting the need for more experiments on feature extraction techniques to better capture their prognostic value. Most importantly, carefully created embeddings are likely to benefit the models.

One key takeaway is that foundation models for histopathology images could significantly enhance feature extraction by identifying the regions of interest better on the images. In this study, histopathology slides were processed using ResNet18 embeddings followed by clustering as in a separate step of the whole model. But more sophisticated models, such as self-supervised vision transformers or

domain-specific histopathology networks, may better capture spatial and morphological patterns in tumor microenvironments. Future work should explore pre-trained histopathology-specific models or fine-tuned large-scale vision models on relevant datasets. For example, UNI is a popular general-purpose self-supervised model for pathology that is pretrained on more than 100 million images from 100,000 diagnostic H&E-stained WSIs including ones from TCGA [23]. Not only does it reduce the preprocessing workload for histopathology images, but it also provides rich, domain-specific feature representations that could improve downstream survival prediction. This allows the network to learn hierarchical representations directly from raw whole-slide images in a fully end-to-end manner.

There is also a strong need for multimodal interpretability of the model. Survival risk prediction is mostly useful in cohort-based treatment and planning. One important question that can be answered using the multimodality approach is the ability to explain the impacts of different data sources [1]. For example, given a prediction on survival risk, we want to attribute how local pixel regions in histopathology images, SHAP values for clinical features, or integrated gradients for RNA-seq data are used to produce that prediction. A simpler approach could be to visualize the softmax attention scores learned from the model in this study. Overall, multimodal interpretability can help identify novel integrative biomarkers with more diagnostic, prognostic, and therapeutic relevance [1].

Lastly, several choices in this model architecture are arbitrarily empirical. More experiments can help with learning these embeddings better. For example, more combinations for the regularizations (dropout rate, weight decay) and training (learning rate, epochs, batch size, number of clusters) can be tested. During training and evaluation, stratifying the data splitting based on event distribution or cross-validation might also lead to more comprehensive results.

The two biggest lessons I learned from this project are (1) the importance of thoroughly understanding the data and designing appropriate embedding strategies with clear rationale, and (2) critically reading research papers and public code to selectively extract only what is necessary for my work.

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